

# Facilitator talking points — Data Analysis in Databricks

Participant notebook: `01-lab.py` Scheduled: 9:45 AM–11:15 AM · 90 minutes

## Delivery map

| Time        | Minutes | Mode                        | Surface and focus                                                           |
|-------------|---------|-----------------------------|-----------------------------------------------------------------------------|
| 9:45–9:50   | 5       | Slides                      | Business question, promotion, FPD5, eligibility, and investigation boundary |
| 9:50–9:57   | 7       | Slides                      | Notebook compute, Jobs compute, and SQL warehouse distinctions              |
| 9:57–10:15  | 18      | Notebook                    | Verify sources; explore application mix and campaign timing                 |
| 10:15–10:27 | 12      | Notebook                    | Build the eligible FPD5 population and compare cohorts                      |
| 10:27–10:39 | 12      | Notebook exercise           | Change analytical grouping and write the handover note                      |
| 10:39–10:49 | 10      | Notebook                    | Persist the result and inspect Delta history                                |
| 10:49–10:56 | 7       | Facilitator demo            | Metric View and prepared AI/BI Dashboard                                    |
| 10:56–11:06 | 10      | Guided participant activity | Create and validate each participant's private Genie Agent                  |
| 11:06–11:11 | 5       | Facilitator demo            | SQL Query History and Query Profile                                         |
| 11:11–11:15 | 4       | Notebook and discussion     | Ownership, operational risk, validation, and recovery                       |

## 1. Business question and FPD5

- Nova Mobile subsidizes selected smartphone loans so customers pay 0% monthly interest; customers still repay principal.
- The question is whether the promotion cohort has higher FPD5 and where the signal is concentrated.
- A contract is eligible only after its first installment reaches day five past due by `2026-09-01`.
- It is FPD5 when the first installment remains unsettled at that point or was settled on or after day five.
- Compare both counts and rates. A high rate is an investigation signal, not proof of fraud or causality.

## 2. Notebook exploration and compute

- Databricks notebooks combine Markdown, Python, PySpark, and SQL in one workflow.
- In this lab, Python, PySpark, `spark.sql(...)`, and `%sql` all run on the notebook's attached Serverless compute.
- `%sql` changes the notebook cell language; it does not move that cell to a SQL warehouse.
- Databricks provides the `spark` session. Participants do not create `SparkContext` or `SparkSession`.
- A SQL warehouse runs the dashboard and Genie Agent queries. Those statements appear in SQL Query History.
- For scheduled Python or notebook work, use Jobs compute rather than keeping interactive notebook compute running.

## 3. Establish the business context

- Start with `loan_application` to understand channels, product mix, stores, and campaign timing.
- Application volume shows demand and concentration; it does not show repayment behavior.
- The busiest 20% of stores handle about 81% of applications. That is not automatically an FPD5 hotspot.

## 4. Build the eligible FPD5 population

- Move from applications to contracts and first installments.
- The denominator is eligible contracts, not every application, approval, or only defaulted contracts.
- Compare the promotion cohort with other eligible originations before changing the analytical grain.
- Reference result: approximately **42.26%** for the promotion versus **21.03%** for other eligible originations.

## 5. Investigate concentration

- Rank segments by FPD5 rate while retaining eligible and FPD5 contract counts.
- Require at least 10 eligible contracts for store or store-associate rankings.
- Participants change the grouping dimension and record the evidence, denominator, caveat, and follow-up question.

## 6. Persist the handover in Delta

- Save the participant's result only after it has a clear handover purpose.
- The participant-specific table avoids collisions in the shared lab schema.
- Delta history shows the original write and the later schema change that adds `review_note`.
- Time travel depends on retained files; it is useful for diagnosis but is not a backup.

## 7. Reuse governed metrics

- `fpd_analysis` contains one row per eligible contract.
- `fpd_metrics` centralizes the shared FPD5 measures and dimensions.
- The notebook, dashboard, and Genie Agent reuse the same Metric View instead of rebuilding the formula.
- Reconciliation means checking that each surface returns the same governed result.

## 8. Dashboard and Genie Agent

- **Dashboard:** distinguish editable draft from published snapshot; viewers use **Individual data permissions** in this workshop.
- **Genie Agent:** it is private to the participant and uses only `fpd_metrics`.
- Inspect generated SQL before trusting the written answer: source, `MEASURE(...)`, denominator, filters, date, and caveat must be correct.

## 9. Operate the SQL workload

- Use Query History to locate dashboard- or Genie-generated SQL warehouse statements.
- In Query Profile, distinguish waiting time, execution time, result fetching, scanned data, pruning, and spill.
- Queueing suggests a concurrency or workload-isolation question; spill suggests reviewing data volume, joins, aggregations, and warehouse sizing.

## Closing question

For each asset created or used, ask: who owns it, how is it validated, where is it monitored, and what is the recovery action?

# Facilitator talking points — Generative AI in Databricks

Participant notebook: `02-lab.py` Scheduled: 11:15 AM–12:15 PM · 60 minutes

## Delivery map

| Time        | Minutes | Mode                     | Surface and focus                                               |
|-------------|---------|--------------------------|-----------------------------------------------------------------|
| 11:15–11:20 | 5       | Slides                   | Genie Code's role and the developer's validation responsibility |
| 11:20–11:27 | 7       | Notebook                 | Use Genie Code to explain the inherited governed SQL            |
| 11:27–11:39 | 12      | Notebook exercise        | Extend the promotion analysis to regional grain                 |
| 11:39–11:44 | 5       | Notebook                 | Run the independent governed regional checkpoint                |
| 11:44–11:54 | 10      | Notebook exercise        | Diagnose and repair the PySpark denominator bug                 |
| 11:54–11:59 | 5       | Notebook                 | Review <code>/optimize</code> and <code>/doc</code> proposals   |
| 11:59–12:08 | 9       | Facilitator demo         | Build and validate the unpublished AI/BI Dashboard draft        |
| 12:08–12:12 | 4       | Optional guided activity | Improve the participant's private Genie Agent, or use as buffer |
| 12:12–12:15 | 3       | Notebook and discussion  | Owner, risk, recovery, and practical next use                   |

## 1. What Genie Code contributes

- Genie Code helps a developer understand, extend, debug, document, and optimize inherited notebook work.
- It can propose edits and execution, but the participant reviews the plan, diff, source, and result before approval.
- The developer remains responsible for the population, denominator, business meaning, permissions, and validation.

## 2. Understand inherited SQL before changing it

- The starting query uses the governed `fpd_metrics` Metric View.
- One row represents one promotion cohort.
- `MEASURE(...)` invokes governed measures; `GROUP BY ALL` groups by the selected dimensions.
- Multiplication by 100 changes display units, not the underlying metric definition.
- Ask Genie Code to explain the query first rather than immediately rewriting it.

## 3. Extend the investigation to region

- Keep `fpd_metrics` as the only source and filter to the promotion cohort.
- Group by `region_code`; retain eligible contracts, FPD5 contracts, and FPD5 rate.
- Apply the 20-eligible-contract rule at regional grain.
- The threshold stabilizes this workshop comparison; it is not part of FPD5 or a statistical-significance claim.

## 4. Validate generated analysis

- Generated code is a draft until an independent query produces the same rows, counts, and rates.
- The governed checkpoint should return six qualifying regions.
- `REGION_VI` is approximately **53.95%**, or **11.69 percentage points** above the **42.26%** promotion baseline.
- If results differ, inspect the source, promotion filter, grouping, measures, `HAVING` rule, denominator, and percentage conversion before continuing.

## 5. Diagnose the PySpark denominator bug

- Let participants observe the suspicious result: both cohorts appear to be 100%.
- The helper filters to `fpd5_flag = 1` before aggregation, removing non-FPD5 eligible contracts from the denominator.
- The repair keeps the full eligible population, counts contracts for the denominator, and sums `fpd5_flag` for the numerator.
- Validate exactly two cohorts, positive eligible counts, FPD5 counts no greater than eligible counts, and Metric View agreement within 0.05 percentage points.

## 6. Notebook compute versus SQL warehouse

- In this lab, Python, PySpark, `spark.sql(...)`, and `%sql` run on the notebook's Serverless compute.
- Genie Code assists with the notebook, but approved notebook cells still run on that notebook compute.
- The AI/BI dashboard and private Genie Agent execute their analytical SQL on a SQL warehouse.
- Notebook `%sql` statements should not be presented as SQL warehouse Query History activity.

## 7. Review optimization and documentation

- `/optimize` is a proposal, not proof that a query is faster.
- Accept a change only when the result remains identical and there is a material rationale.
- Performance claims require Query Profile evidence.
- `/doc` should capture the grouping, eligible-contract denominator, and threshold without duplicating the FPD5 formula.

## 8. Turn evidence into a dashboard

- The disposable dashboard uses only `fpd_metrics` and remains unpublished.
- KPI counters show the all-cohort baseline.
- The promotion filter affects the cohort comparison and store ranking, not the all-cohort KPI counters.
- Store rankings retain eligible and FPD5 counts beside the rate and require at least 10 eligible contracts.
- The point is to show that generated presentation assets still require reconciliation and publication decisions.

## 9. Optional private Genie Agent improvement

- Continue with the participant's private Section 01 Agent; do not create a second Agent.
- Keep `fpd_metrics` as its only data source.
- Add the smallest instruction or verified example needed for counts, observation date, investigation language, and the 10-contract ranking rule.
- Test the target question and the original cohort comparison in fresh conversations to detect regression.

## Closing question

Where did AI accelerate the work, and which review or validation step prevented speed from weakening trust?

# Facilitator talking points — Governance and Access Control

**Participant guide:** `05-participant-guide.md` **Facilitator demo:** `facilitator-demo.py` **Scheduled:** 3:20 PM–3:50 PM · 30 minutes

## Delivery map

| Time      | Minutes | Mode                             | Surface and focus                                                  |
|-----------|---------|----------------------------------|--------------------------------------------------------------------|
| 0:00–0:03 | 3       | Slides                           | Access incident, assignment, and smoke-test boundary               |
| 0:03–0:07 | 4       | Slides                           | Authorization chain and evidence to collect                        |
| 0:07–0:16 | 9       | Facilitator demo                 | Restricted-user failure, administrator repair, and unchanged rerun |
| 0:16–0:20 | 4       | Slides and discussion            | What the successful repair proves and does not prove               |
| 0:20–0:24 | 4       | Facilitator demo                 | Table and column lineage in Catalog Explorer                       |
| 0:24–0:27 | 3       | Facilitator demo                 | Consumer Lending Domain in Discover                                |
| 0:27–0:30 | 3       | Participant guide and discussion | Complete the incident handover                                     |

## 1. Access is a chain of independent gates

Use the incident:

A risk analyst can discover `hc_workshop.workshop_shared.fpd_analysis`, but cannot query it.

The request must pass:

Open notebook → use SQL warehouse → identify runtime identity  
→ USE CATALOG → USE SCHEMA → SELECT → return authorized result

Passing one gate does not prove the next gate will pass.

## 2. Identify the executing identity

- A user, service principal, or group can hold permissions.
- The runtime identity is the user or service principal executing the request.
- Groups contribute effective access through membership; the group is not the interactive user returned by `current_user()`.
- A scheduled Job may use a different **Run as** identity from the person who configured it.
- Confirm the runtime identity before changing grants.

## 3. Separate workspace access from data access

- Notebook **CAN RUN** controls whether the identity can execute the workspace asset.
- SQL warehouse **CAN USE** controls whether it can submit work to that warehouse.
- Unity Catalog privileges control access to the governed data object.
- Opening the notebook and using its warehouse do not prove the view can be queried.

## 4. Explain the Unity Catalog read path

- **USE CATALOG** allows traversal into `hc_workshop`.
- **USE SCHEMA** allows traversal into `workshop_shared`.
- **SELECT** allows rows to be read from `fpd_analysis`.
- **BROWSE** supports discovery and metadata visibility; it does not grant row access.
- Privileges may be direct, inherited from a parent securable, or received through group membership.

## 5. Diagnose and repair the incident

- The restricted user can run the notebook, use the warehouse, discover the asset, pass `USE CATALOG` , and has `SELECT` .
- The unchanged query fails because `USE SCHEMA` is missing.
- Record the exact error, request ID, runtime identity, object, action, compute, and permission evidence.
- Grant only `USE SCHEMA` to the dedicated restricted group.
- Rerun the same SQL with the same user, warehouse, and object.
- Ten returned rows verify this access repair; they do not validate the FPD5 business definition or prove unrestricted row visibility.

## 6. Apply the narrowest repair

- Preserve the failing request and change one authorization variable at a time.
- Do not use `ALL PRIVILEGES` as a troubleshooting shortcut.
- Do not grant access to the person reporting the error until the executing identity is known.
- A successful unchanged rerun provides stronger diagnostic evidence than changing the query and the permission together.

## 7. Use lineage for impact analysis

- Lineage shows upstream sources and downstream dependencies.
- Follow `fpd_analysis` → `fpd_metrics` → `verified dashboard or analysis` .
- Use it to decide which assets require retesting after a definition, schema, access, or availability change.
- Lineage does not grant access, prove correctness, or replace audit evidence.

## 8. Separate discovery from authorization

- **Discover Domains** organize trusted assets by business purpose.
- **Consumer Lending > Origination Risk** helps consumers find `fpd_analysis` .
- Domain placement proves curation for discovery; it does not grant `USE CATALOG` , `USE SCHEMA` , or `SELECT` .
- Use Discover for business navigation and Catalog Explorer for detailed metadata, permissions, and lineage.

## Closing handover

Participants should be able to state the runtime identity, failed gate, evidence, narrow repair, unchanged verification, downstream assets to retest, Domain boundary, owner, risk, and recovery action.